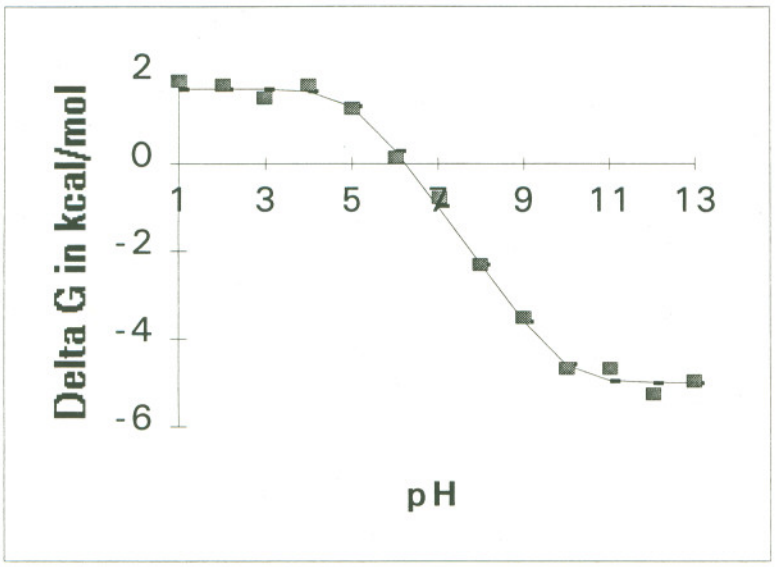
Biochemistry Cumulative Examination
September 18, 1993

You are investigating a system where one compound, N, is interconverting with another compound, D. You can measure the free energy difference between compound N and compound D in the reaction N → D. You do so at a variety of pHs and find that the free energy difference between these two compounds shows a pronounced pH dependence. This data is summarized below. You know from spectral and chemical evidence that there is a single ionizable group in both N and D but due to changes in the environment of this group its pK_a shifts.



pH	ΔG for N → D
1	1.9
2	1.8
3	1.5
4	1.78
5	1.25
6	0.17
7	-0.78
8	-2.31
9	-3.5
10	-4.7
11	-4.69
12	-5.24
13	-4.98

1. You believe that this group is an amine. Does the data presented above allow you to prove or disprove this? Explain *briefly*. 10 points.

2. What is the usual pK_a range you would expect to see for an amine? 5 points.

3. Continuing with the premise that the ionizable group is an amine and using the notation N^+, N^o, D^+ and D^o draw the thermodynamic cycle that links these species. 10 points.

4. Let the free energy difference between N and D be ΔG_{pH}. The subscript is to indicate that this free energy difference is pH dependent. Write the equation that relates the value of ΔG_{pH} to the concentrations of N^+, N^o, D^+ and D^o. 15 points.

5. What is the equilibrium relationship between D^+ and D^o at a given pH? Hint: The Henderson-Hasselbalch equation is $pH = pK_a + \log [A^-]/[HA]$. 15 points.

6. Give an equation for the free energy difference between N and D (overall) as a function of pH. Hint: The equations you wrote in 4 and 5 may be of some use. 25 points.

7. This is not merely an abstract exercise. Protonation/deprotonation events are energetically coupled to many biological processes. Give an example of energetic coupling to a protonation/deprotonation event. 10 points.

8. Examine the equation you have derived in 6 and the data above. Assume that the interconversion of N and D is not directly pH dependent (i.e. the free energy difference between N^+ and D^+ is a constant as is that between N^o and D^o) but rather that the pH dependence exhibited above is linked to the shift in pK_a of the group you think is an amine. Do the data support this assumption? Why or why not? Give a rough estimate of the energy difference between N^o and D^o at pH 13. At pH 3. Give a rough estimate of the pK_a of the putative amine in the two species N and D. 10 points.